The Slow Architecture of Coral

Student copy (project or hand out)

1 A coral reef is built out of a mineral the animal manufactures one tiny particle at a time. The mineral is aragonite, a form of calcium carbonate, the same compound as chalk but a different crystal arrangement than the calcite that makes up most limestone. A polyp does not pull finished crystal from the sea. It assembles soft, formless beads of calcium carbonate first, each about 400 nanometers across, and holds them in its tissue for hours before they harden into aragonite. That detour through a formless phase is roughly a hundred times faster than growing the crystal ion by ion straight out of seawater.

2 To make the mineral at all, the polyp has to change the chemistry of a sliver of space. Between its own cells and the skeleton it has already laid down, it pumps hydrogen ions out of that gap. Removing the hydrogen tips the water's balance toward carbonate ions, and once enough carbonate has gathered there, calcium and carbonate can lock together into solid calcium carbonate. A separate set of proteins, secreted by the coral and called the soluble organic matrix, folds itself into ordered layers that steer which crystal forms, hold the formless beads stable, and suppress rival minerals the chemistry would otherwise allow. The skeleton grows the way it does because the animal keeps editing the conditions under which it grows.

3 None of this would sustain a reef without a tenant. Inside the polyp's tissue live single-celled algae, photosynthetic dinoflagellates called zooxanthellae, packed at more than a million cells in every square centimeter. The coral breathes out carbon dioxide and water; the algae take both and run photosynthesis; and the algae hand as much as ninety percent of what they make back to the coral that houses them. The lipids alone tell the story of the bargain: of the wax esters and fats the algae produce, about seventy-five percent goes to the host and only eight percent stays behind. It is an arrangement that lets coral flourish in tropical water so nutrient-poor it should be close to a desert.

4 Light feeds the skeleton as well as the animal. When the algae photosynthesize they pull carbon dioxide out of the surrounding tissue, the local water turns less acidic, carbonate becomes easier to come by, and the coral lays down skeleton faster in the light than in the dark. The effect has a name, light-enhanced calcification, and for a long time the obvious reading was that the algae simply fed the process. The picture turns out to be less tidy. Under blue light, calcification can hold near its normal pace even when the algae's photosynthesis is turned down, which points to blue-light receptors in the coral itself triggering the building through some signal we do not yet fully trace.

5 The building runs on a clock, or several clocks at once. Through the day the algae convert sunlight into chemical energy; through the night the colony shifts toward feeding on the water and toward repair. Many stony corals open and close their polyps on a daily beat set by light, and time their spawning to particular phases of the moon. The skeleton itself keeps the time. Its fibers are laid down in rhythmic pulses, repeated

deposits of acidic proteins building a layered structure pulse on pulse, the way a tree lays a ring. Genus by genus the pace differs, somewhere between a third of a centimeter and two centimeters a year, depending on warmth and food and how clear the water is.

6 Because that layering is steady and the chemistry records its surroundings, a single old colony becomes an archive. Mounding corals of the genus Porites can live for centuries; their annual density bands store, in order, the seawater temperature and salinity and acidity they grew through, month by month and year by year. Cores drilled from reefs reach back across the ten thousand years of the Holocene. There is a limit worth keeping in view. The same warmth that the bands faithfully record can also break the partnership the colony is built on: pushed past the hottest water it normally meets, a coral will digest and cast out the very algae that feed it, and the long patient record goes on being written by the thing that can end it.

AP-style multiple choice:

1. In the fourth paragraph, the writer follows the explanation of light-enhanced calcification with the observation that 'The picture turns out to be less tidy' primarily to

- A) concede that the role of light in building the skeleton remains entirely unknown
- B) argue that earlier researchers misunderstood how coral and algae cooperate
- C) qualify the straightforward account by introducing a finding the simple version does not explain
- D) emphasize that blue light is the most important factor in coral growth

2. In context, the writer's choice to call the old colony 'an archive' whose bands 'store' temperature and salinity 'in order' best conveys which of the following about the writer's stance toward the coral skeleton?

- A) that the writer regards the reef chiefly as a resource for human study
- B) that the skeleton performs an orderly, deliberate-seeming act of keeping a record over time
- C) that the coral consciously intends to document its environment for the future
- D) that the skeleton is fragile and its record easily erased

Discussion questions:

1. The piece spends its first two paragraphs on chemistry the reader cannot see (formless beads, pumped hydrogen ions, steering proteins) before any mention of the algae most people associate with coral. What does the writer gain by building the skeleton's chemistry first and introducing the famous partnership only afterward?

2. The closing sentence notes that the same warmth the skeleton faithfully records is the warmth that can make a coral expel the algae that feed it. How does ending on that doubling, rather than on a summary of how coral builds, change what the reader is left holding?

Writing prompt (Homework or extended in-class, Q2 rhetorical analysis):

In the passage above, the writer describes how coral constructs its skeleton and sustains its partnership with the algae that live inside it. Read the passage carefully. Then write an essay in which you analyze the rhetorical choices the writer makes to convey the precision and scale of coral as a biological system. Support your analysis with specific evidence from the text, attending to choices such as the ordering of information, the movement between short declarative sentences and long accumulating ones, figurative language, and the writer's candor about what is not yet fully understood.